

AJAY VENKATESH

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Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Sep 2024 – May 2026

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Skills

Programming Languages: Python, Java, C++, TypeScript, JavaScript (ES6), SQL

Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize

Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse

Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs), Helm

Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight, Prometheus, Grafana, CloudWatch

AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark

Experience

Campus Mesh

Jan 2026 – Present

Software Engineer | Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines

Developed and maintained scalable Node.js microservices powering real-time academic workflows for 10K+ users, including study groups, messaging, notifications, and profile management.

Built a personalized recommendation engine using collaborative filtering and behavioral analytics, increasing user engagement by 28% and retention by 50%.

Authored LLM-based retrieval pipelines for CampusIQ (AI Assistant), improving contextual accuracy by 30% through semantic retrieval and structured data indexing.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern | Serverless AWS infrastructure, React 18 frontend, and Java backend systems

Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the system design for approval and established Role Based Access (4+ roles) using OAuth, Midway SSO, and HELIS.

Provisioned serverless Infrastructure-as-Code using AWS CDK with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by 70% and enabling multi-stage, multi-region CI/CD deployments.

Built a React 18 frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by 40%.

Engineered backend services in Java using Coral, Smithy, Dagger, and CBOR serialization; applied low-level design and design patterns reduced backend execution time by 35% and ensure reliability by unit and integration testing using Mockito.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus) | NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics

Architected an NSF-funded (\$100K) AI-powered health app for iOS and Android, processing real-time wearable data; deployed cloud-hosted AI models for health matrix analysis and process food images.

Engineered ETL pipelines transforming NoSQL into PostgreSQL using concurrency control and locking mechanisms; processed 250K+ records and powered AWS QuickSight dashboards for real-time health analytics.

Developed an insurance analytics platform (React, Node.js, REST APIs); deployed (Docker containers on EC2, implemented Stripe billing, RBAC, and row-level security, generating \$50K+ revenue.

Streamlined underwriting via broker portal integrated with insurer platform; implemented document and media processing and WebSocket-based real-time communication; secured 6 insurance clients.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer | Cloud-native backend, NLP/OCR automation, and data engineering on Azure

Developed a distributed backend on Microsoft Azure for a health platform, integrating with Microsoft CRM and implementing event-driven processing and fault tolerance.

Led code reviews and mentored engineers on software design principles, reducing production bugs by 25%.

Deployed an NLP based OCR neural model on Azure to automate structured data extraction, reducing data processing time by 75%.

Optimized data ingestion with parallel processing for 100K-row from Excel; reduced execution time from 5 hours to 1 hour.

Built Power BI dashboards processing large datasets from multiple databases, improving reporting efficiency by 40%.

Projects

Personal Finance Analyzer (Python Data Analysis Project) | Python, Pandas, NumPy, Matplotlib, Seaborn

Built a data-driven application to analyze personal financial transactions and categorize expenses using Pandas for efficient data processing.

Generated interactive visualizations and spending insights using Matplotlib and Seaborn to identify trends and optimize budgeting.

Designed reusable and modular code structure following OOP principles to improve maintainability and scalability.

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Microsoft Solution Architect Expert

Power BI Data Analyst Associate

Microsoft Developer Associate

Stanford Machine Learning (Coursera)